* she's going rlly fast... check slides (I'm writing what she's saying mostly)

Lecture notes for Keigster 1/21/26

Operators: $\hat{O}|x\rangle = |x'\rangle$

- $|x\rangle$ not necessarily normalized
- $\hat{O}$ is a matrix, treating kets as vectors (Hilbert space)

Adjoints: $\hat{O}^\dagger$ operates on bras

- Transpose and complex conjugate: $\hat{O}^\dagger = \begin{pmatrix} O_{11}^* & O_{21}^* \\ O_{12}^* & O_{22}^* \end{pmatrix}$

$O_{jk} \equiv \underbrace{\langle j|\hat{O}|k\rangle}_{\text{basis vectors}}$ extract element/define CE → take j, k basis vectors, do $\hat{O}|k\rangle$ then $\langle j|$ on it

Hermitian Operators: represent things we can actually observe in nature

- encode possible measurable outcomes → need real quantities
- Real λ's (physical outcomes) + $\vec{v}$'s ← eigenvector complete, orthonormal
  - all possible outcomes are basis of HS

Can always express $\hat{H} = \sum_{n=1}^{N} \lambda_n |n\rangle\langle n|$ → $\begin{pmatrix} n \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ n \end{pmatrix}$ → $n \times n$ matrix? (check this)

N eigenvectors eigenvectors

$\hat{H}|\psi\rangle \to \sum_{n=1}^{N} \lambda_n |n\rangle\langle n|\psi_n\rangle$

Check if Hermitian: write operator, λ_n real, can $(|n\rangle\langle n|)^\dagger = |n\rangle\langle n|$

so $\hat{H} = H^\dagger$ (trans, CC, see if same as original)

- operator in ~~diagonal~~ natural basis: diagonal matrix

Unitary Operators: normalized ket = good QS → preserves length^2 to describe state

$\langle\psi|\psi\rangle = \langle\psi'|\psi'\rangle = 1$? (normalized)

$\langle\psi|\hat{U}^\dagger\hat{U}|\psi\rangle$ · Check unitary: multiply $\hat{U}$ by $\hat{U}^\dagger$ adjoint

acts on

Hermitian encodes observables, doesn't collapse state

Projection operator: $|\phi\rangle\langle\phi|$ = Projection operater (matrix) on

- Lose info from original state
- on $|+x\rangle$ expressed in z-basis

Generic Change of Basis: $|1\rangle\langle 1|x\rangle$ → coefficient

$= \underbrace{\left(\sum_{n=1}^{N} |n\rangle\langle n|\right)}_{\mathbb{I}}|x\rangle$ * physicists love this cool trick

$\mathbb{I}$ identity operator: sum over projections on every basis element

inserting a basis/complete set of states

express in some basis

choose any complete orthonormal basis → need to

new basis decompose ket

1st basis → insert $\mathbb{I}$ (primed basis) → switch sum → inner product is linear complete/orthonormal

$\langle n'|n\rangle$, then new $|n'\rangle$ ket

number

all expressions in old basis × change of basis elements

Parentheses is new CE

L5Q3: $|+_z\rangle\langle +_z| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Hermitian: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}^{\dagger} \overset{\text{trans}}{=} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \overset{cc}{=} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ ✓

Unitary: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ ✗ not I matrix

- unitary needs adjoint inverse (hint: $\lambda_2 = 0$)

- Projection never unitary
- Sum over all proj. for all basis elements? Get identity = unitary

Lecture 7: Quantum Uncertainty

$\sum_{n=1}^{N}$ sum over all directions

- $\hat{M}|\Psi\rangle$ producing observables since m_n has units

$\langle m_n|, |m_j\rangle$ basis vectors so Kronecker-delta applies

LSQ4: $\hat{M}|\lambda\rangle = \lambda|\lambda\rangle$

* Practice: verify S_x and S_y (not diagonal... expressed in z-basis)

spin $\frac{1}{2}$ example:

- $\pm\frac{\hbar}{2}$ units of angular momentum

S_z: $\frac{\hbar}{2}|+_z\rangle\langle +_z| - \frac{\hbar}{2}|-_z\rangle\langle -_z|$? I think (yes)

Multiply out matrices

Vector spin operator: $\xrightarrow{}$ 2x2 matrix x Euclidean basis vector

$\hat{\vec{S}} \cdot \vec{e}_r$ = spin in e_r dir'n (generic)

* specialize θ, ϕ for $\hat{S}_x, \hat{S}_y, \hat{S}_z$ *do eigenvalues for this and eigenvectors

$|+_r\rangle$ generic spin state "pointed" in dir'n (θ, ϕ)